Project Management Methodologies

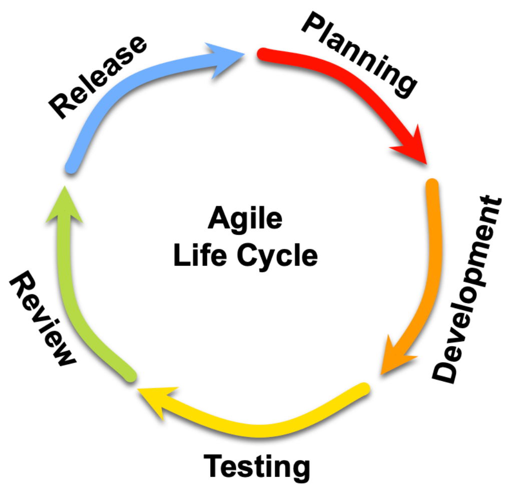
Agile Methodology

Agile is a project management approach used in software development that focuses on flexibility, collaboration, and incremental progress. Rather than attempting to define every requirement at the beginning of a project, Agile encourages teams to deliver small pieces of working functionality and continuously refine the product as new feedback becomes available. This allows development teams to adapt to changing requirements and improve the product throughout the development lifecycle (Beck et al., 2001).

In Agile development, work is typically organised into repeated cycles of planning, development, testing, and review. These cycles allow teams to evaluate progress regularly and adjust priorities where necessary. Agile approaches also emphasise communication between team members and encourage close collaboration when making design and implementation decisions.

Agile has become widely adopted in modern software development because it enables teams to respond to uncertainty and evolving requirements more effectively than traditional sequential approaches (Agile Alliance, n.d.).

Example Agile lifecycle



1. Planning
   During the planning stage, the team reviews project priorities and determines which features or tasks should be addressed next. Requirements are clarified, tasks are broken down into smaller units of work, and responsibilities are assigned.
2. Development
   In the development stage, team members implement the planned functionality by writing code, creating user interface components, and integrating system features according to the project requirements.
3. Testing
   Testing involves verifying that the implemented functionality behaves as expected and does not introduce errors. This may include manual testing, automated testing, and checking that different parts of the system work together correctly.
4. Review
   During the review stage, the completed work is evaluated to ensure it meets the intended requirements and quality standards. This may involve peer code review, validation of the implemented feature, and identifying improvements before release.
5. Release
   The release stage involves making the completed functionality available within the application or preparing it for deployment. Once released, feedback can be gathered and used to inform the next planning phase.
6. (Repeat)
   Because Agile development is iterative, this cycle repeats throughout the project, allowing the team to continuously refine the system and respond to feedback or changing requirements.

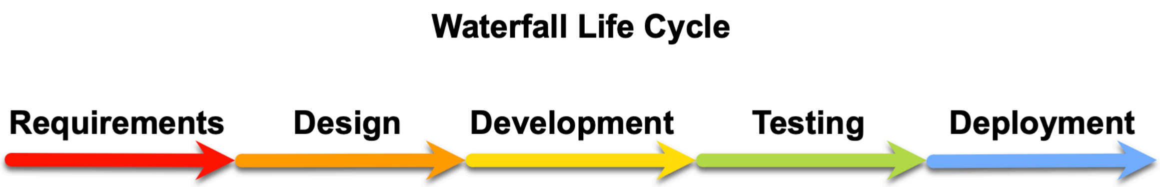
This iterative process supports continuous learning and gradual improvement of the system being developed.

Waterfall Methodology

Waterfall is a traditional project management methodology that follows a sequential and structured process. In a Waterfall model, each stage of the project is completed before the next stage begins. The typical phases include requirements gathering, system design, implementation, testing, and deployment (Atlassian, n.d.).

In this model, the majority of planning and design work occurs at the beginning of the project. Once development begins, changes to earlier decisions can be difficult or costly to implement because later stages depend on the completion of earlier ones.

Example Waterfall workflow



1. Requirements
   During the requirements stage, the goals and functional requirements of the system are identified and documented. Stakeholder needs are analysed to determine what the system must achieve before development begins.
2. Design
   In the design stage, the system architecture and technical structure are planned based on the documented requirements. This may include designing the database structure, application architecture, and user interface layout.
3. Development
   During development, the planned system components are implemented by writing the necessary code and building the application according to the design specifications.
4. Testing
   Testing is conducted to verify that the system functions correctly and meets the original requirements. This stage involves identifying and resolving bugs, as well as ensuring the system behaves as expected.
5. Deployment
   In the deployment stage, the completed system is released for use. This may involve installing the software on a production environment or making the application available to end users.

Waterfall can be effective for projects where requirements are well defined and unlikely to change. However, it may be less suitable for projects where requirements evolve during development or where experimentation and iteration are beneficial (Atlassian, n.d.).

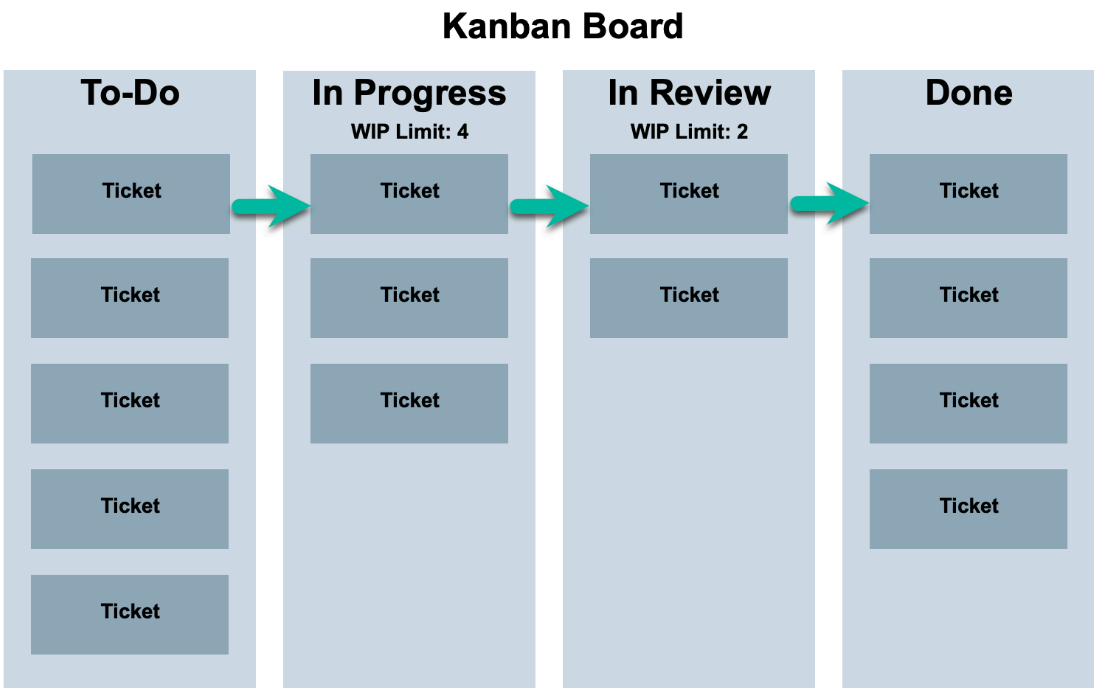
Task Management Methodologies

Kanban

Kanban is a workflow management method used to visualise and manage tasks as they move through different stages of completion. Work items are typically represented as cards on a board, which move across columns representing stages such as backlog, in progress, and completed.

The main objective of Kanban is to improve the flow of work by making tasks visible and limiting the amount of work being performed at any one time. These limits, known as Work-In-Progress (WIP) limits, help teams avoid overloading themselves and encourage completion of tasks before new ones are started (Anderson and Carmichael, 2016).

Example Kanban workflow



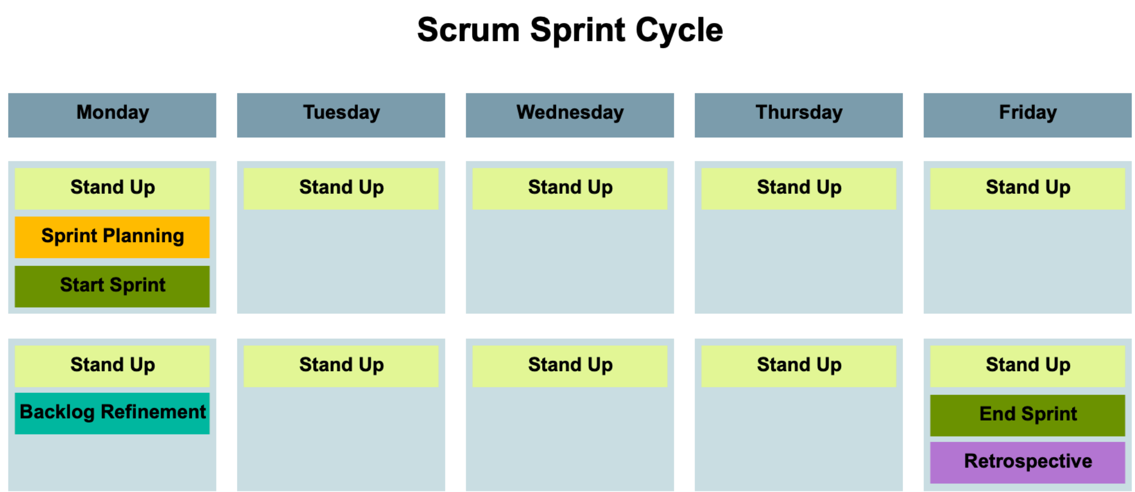
Each task is represented as a ticket that moves across workflow stages, beginning in the "To Do" column and progressing through stages such as "In Progress", "In Review", and "Done". Work-in-progress (WIP) limits are applied to certain stages to reduce bottlenecks and ensure that team members focus on completing tasks before starting new ones.

Kanban supports continuous delivery because tasks can be completed and released individually rather than waiting for a fixed development cycle.

Scrum

Scrum is another Agile framework that organises work into structured development cycles called sprints. Each sprint typically lasts between one and four weeks and includes several defined activities such as sprint planning, daily meetings, sprint review, and retrospective (Schwaber and Sutherland, 2020).

Example Scrum sprint cycle



At the beginning of the sprint, the team conducts Sprint Planning to select tasks from the product backlog and define the work that will be completed during the sprint. Daily stand-up meetings are held throughout the sprint to allow team members to briefly share progress, identify obstacles, and coordinate work. During the sprint, activities such as backlog refinement may also occur to clarify and prioritise upcoming tasks. At the end of the sprint, the team completes the work cycle with a Sprint Review and

Retrospective, where the completed work is evaluated and the team reflects on ways to improve the development process for the next sprint.

Scrum introduces defined roles, including the Product Owner, Scrum Master, and Development Team, to support communication and coordination during the development process.

Methodology Selection for this Project

Project Management Approach

For the development of The Village, the team has chosen to adopt an Agile project management approach rather than a Waterfall model.

Agile is well suited to this project because the application is being developed in stages across multiple assessments, including planning, frontend development, backend implementation, and deployment. Using an Agile approach allows the team to progressively build the system while refining features and priorities based on ongoing discussions and feedback.

Additionally, all team members are balancing this project alongside other commitments. The flexibility offered by Agile enables the team to adapt planning and task allocation based on availability while still maintaining steady progress.

Task Management Approach

To manage day-to-day development tasks, the team is using Kanban through a Jira board.

Kanban was selected instead of Scrum because it allows tasks to move continuously through the workflow without requiring strict time-boxed sprints. This approach provides greater flexibility for a team working on this project part-time and enables members to begin tasks when they have capacity.

The Jira Kanban board is used to visualise the workflow and track progress across several stages, including “To Do”, “In Progress”, “Review”, and “Done”.

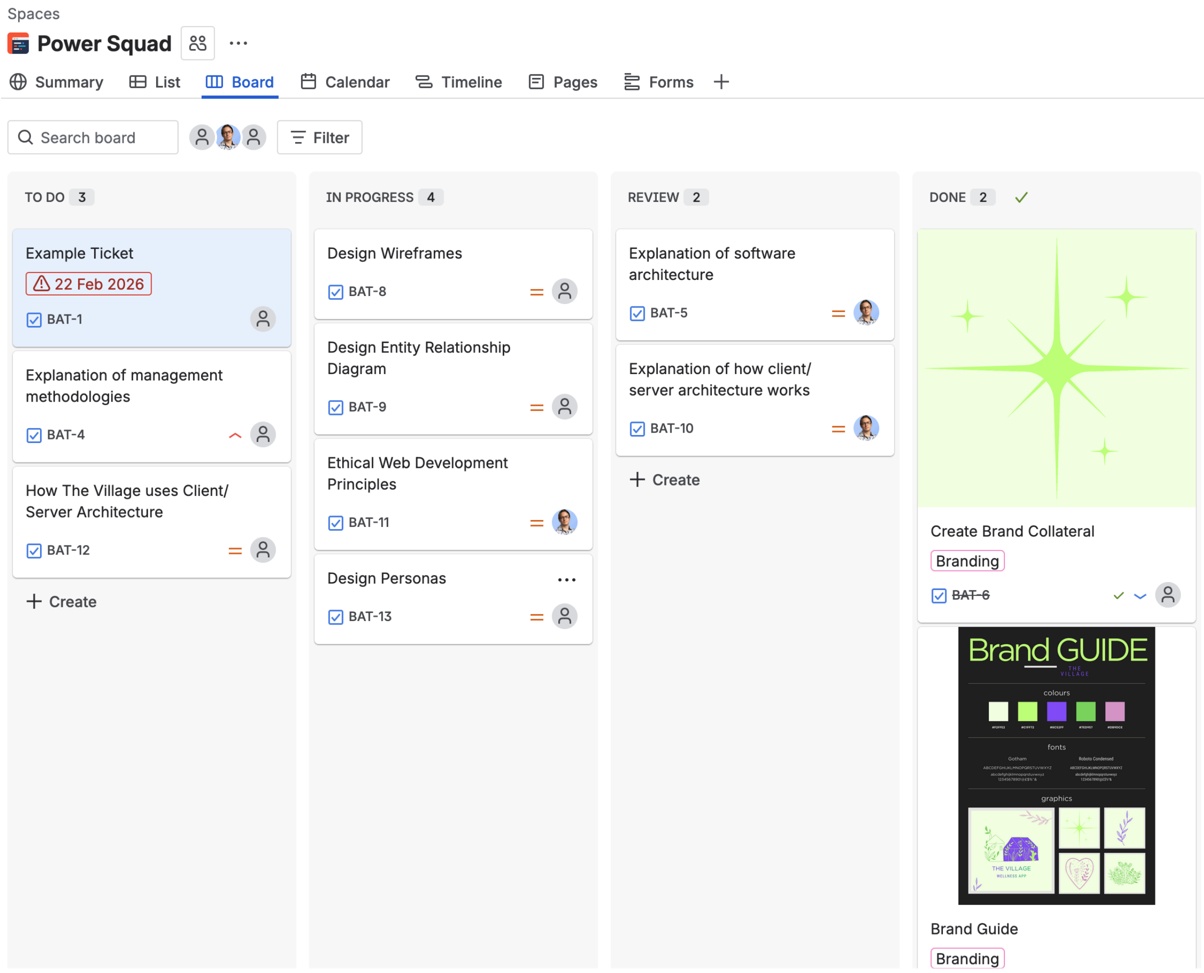
Each task is represented as a Jira ticket and moves across these stages as work progresses. This allows the team to clearly see the current status of tasks and identify any bottlenecks.

Project Planning Tools

Two main tools are used to support collaboration and planning within the project:

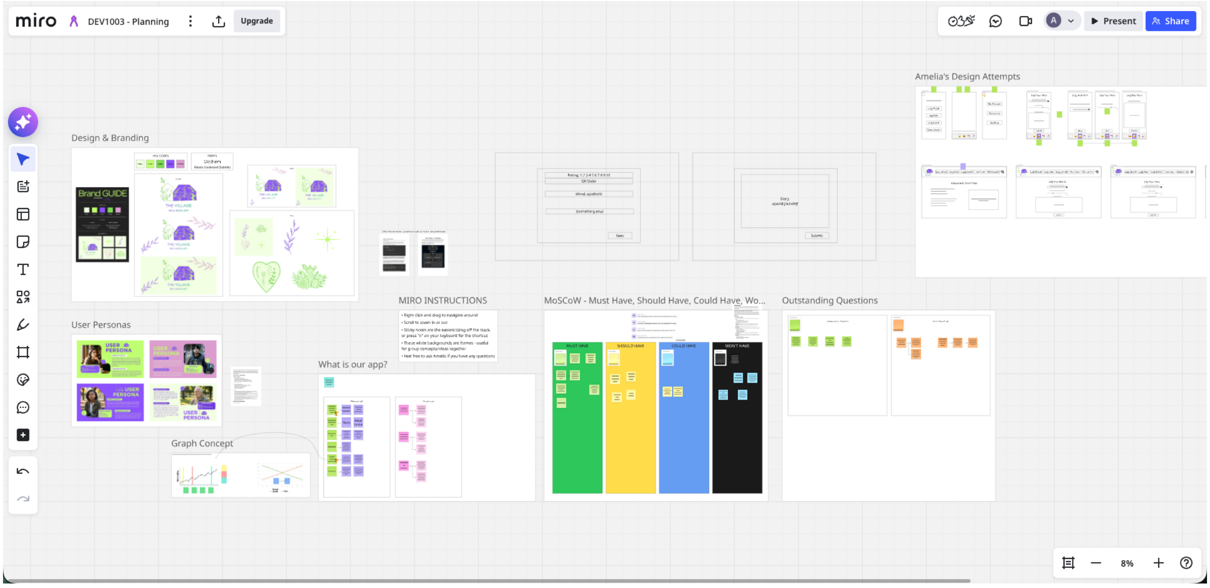
Jira

Jira is used to manage development tasks and organise the workflow using a Kanban board. Tasks are recorded as tickets and assigned to team members, allowing progress to be tracked transparently.



Miro

Miro is used for collaborative planning activities such as brainstorming features, mapping system components, and organising higher-level project ideas before they are converted into actionable Jira tasks.



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